

Translocation of Crohn's disease Escherichia coli across M-cells: contrasting effects of soluble plant fibres and emulsifiers

Background Crohn's disease is common in developed nations where the typical diet is low in fibre and high in processed food. Primary lesions overlie Peyer's patches and colonic lymphoid follicles where bacterial invasion through M-cells occurs. We have assessed the effect of soluble non-starch polysaccharide (NSP) and food emulsifiers on translocation of Escherichia coli across M-cells. Methods To assess effects of soluble plant fibres and food emulsifiers on translocation of mucosa-associated E coli isolates from Crohn's disease patients and from non-Crohn's controls, we used M-cell monolayers, generated by co-culture of Caco2-cl1 and Raji B cells, and human Peyer's patches mounted in Ussing chambers. Results E coli translocation increased across M-cells compared to parent Caco2-cl1 monocultures; 15.8-fold (IQR 6.2–32.0) for Crohn's disease E coli (N=8) and 6.7-fold (IQR 3.7–21.0) for control isolates (N=5). Electron microscopy confirmed E coli within M-cells. Plantain and broccoli NSP markedly reduced E coli translocation across M-cells at 5 mg/ml (range 45.3–82.6% inhibition, $p<0.01$); apple and leek NSP had no significant effect. Polysorbate-80, 0.01% vol/vol, increased E coli translocation through Caco2-cl1 monolayers 59-fold ($p<0.05$) and, at higher concentrations, increased translocation across M-cells. Similarly, E coli translocation across human Peyer's patches was reduced $45\pm 7\%$ by soluble plantain NSP (5 mg/ml) and increased 2-fold by polysorbate-80 (0.1% vol/vol). Conclusions Translocation of E coli across M-cells is reduced by soluble plant fibres, particularly plantain and broccoli, but increased by the emulsifier Polysorbate-80. These effects occur at relevant concentrations and may contribute to the impact of dietary factors on Crohn's disease pathogenesis.